

UCLA TA1 — AIQ Phase 1 Submission Summary

UCLA TA1 Team (VeriStressGT)

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Submission. Evaluation card `cards/thrust1_soundness.yaml`, branch `evaluation-card` of <https://github.com/dtroxell119/VeriStressGT>. Submitted 9/29, ahead of the 10/9 deadline, because the submitter is unavailable for the two weeks after submission; they are back before preliminary results (10/30), so the results-repair window (10/30–11/13) is covered. Setup and run instructions are in `AIQ_README.md` in the repository.

1 Approach

VeriStressGT tests neural-network verifiers on networks whose robustness answer is known by construction, so a wrong verdict is provably wrong. Existing benchmarks such as VNN-COMP have no ground truth and fall back on majority vote between verifiers.

- **Benchmark:** 50 instances, pre-built in the repository. 24 are UNSAT by analytic construction (exact-radius MILP, MEAP, corners, paired-bias CNN, contractive CNN, attention). 26 are SAT, each with a counterexample checked in float64 and onnxruntime float32; 4 of these are attack-hard, so only a completed proof decides them.
- **Verifier pool:** 2 sound in-house controls; 5 real verifiers (α - β -CROWN, PyRAT, nnenum, NeuralSAT, Marabou 2.0.0); and 42 deliberately buggy verifiers: 12 mutants of our reference verifier plus 6 output-level faults on each real verifier.
- **Scoring:** every definitive verdict is judged against the known label. The same verdicts are also judged by majority vote, as a baseline.

2 Metric and BAA alignment

The BAA metric is `gt_scoring_accuracy`: the fraction of all definitive verdicts that ground truth judges correctly, target $\geq 95\%$. It is 100% in our runs, over 2,025 verdicts.

Card symbol	Measures	Target	Result
<code>gt_scoring_accuracy</code> (BAA)	Verdicts judged correctly by ground truth	$\geq 95\%$	100%
<code>gt_detection_rate</code>	Planted bugs that fired and were caught	$\geq 75\%$ (claim)	100%
<code>mv_full_scoring_accuracy</code>	Same verdicts judged by majority vote	baseline	96.7%

- **Distinct models:** the instances span many distinct networks in three families (MLPs, CNNs, attention models), well past the BAA minimum of two.
- **Datasets:** synthetic constructions; the input data is part of each network’s construction.
- **Claim:** the card passes only if ground truth scores 100% of verdicts, flags no sound control, and catches at least 75% of the planted bugs that fired.

3 Expected results

The run should end with **RESULT: VERIFIED**, with both metrics at 1.0 in the top-level `verdict.json`. Our reference run took 32 minutes on a MacBook Pro (Apple Silicon, 8 GB, CPU only) with MAGNET v0.1.0. The card has been run on macOS; Linux x86_64 is the expected evaluation platform.

Buggy verifiers caught (42 planted, 39 fired)

Labels used for scoring	Caught	Sound verifiers wrongly flagged
Ground truth	39/39 (100%)	none
Majority vote, one buggy verifier	37/39 (95%)	none
Majority vote, full pool	39/39 (100%)	3: reference , ibp_only , real NeuralSAT

Scoring accuracy (2,025 definitive verdicts)

Labels used for scoring	Correct	Wrongful accusations	Wrongful acquittals	Ties
Ground truth	2025/2025 (100%)	0	0	0
Majority vote, one buggy verifier	2015/2025 (99.5%)	0	0	10
Majority vote, full pool	1959/2025 (96.7%)	19	25	22

What can differ on other hardware, without changing the verdict or the metrics:

- **Verdict counts:** timeouts depend on machine speed, so totals (2,025 verdicts, 39 fired bugs) can shift slightly. Ground-truth accuracy stays 100% because the labels do not depend on timing.
- **Marabou:** Marabou 2.0.0 is nondeterministic. We saw it return false SAT on `meap_02`, `meap_03` and `milp_s2_a`, and not every run hits all three. Each is a real Marabou soundness bug that ground truth correctly flags.

4 Scalability

TA2 scales the card with one parameter, **scale:** `N`, in the card’s `algo_params`. The runner builds a benchmark `N` times larger with fresh seeds (`thrust1_bench_xN/`) and runs the same verifier pool on it.

- **What grows:** N re-seeded copies of every base network, each keeping its analytic UNSAT certificate, plus N times the random and attack-hard CNN SAT instances.
- **Why results stay consistent:** every label is re-derived from construction and every SAT witness is re-certified, so expected ground-truth accuracy is 100% at any N .
- **Tested:** at scale: 2, 98 instances (48 UNSAT, 50 SAT); all 50 witnesses re-certified, ground-truth scoring 173/173, no control flagged (sound controls only, 9 minutes, free size-limited Gurobi license).
- **Needs:** gurobipy (installed by the setup script) to build larger benchmarks.

5 Supporting evidence

On external benchmarks most planted bugs cannot be caught at all. We ran the same pool on VNN-COMP `mnist_fc` (30 of 90 instances) and `oval21` (all 30), labelled only by attack-certified counterexamples, since nothing outside our constructions certifies robustness.

Benchmark	Bugs that fire	Caught by own labels	Verdicts checkable
VeriStressGT	39/42	39/42	100%
VNN-COMP <code>mnist_fc</code> (30/90)	19/42	6/42	21%
VNN-COMP <code>oval21</code>	19/42	2/42	7%

- **Real bug found:** unmodified Marabou 2.0.0 answers UNSAT on `oval21` instance `vb_oval21_0014`, which a certified counterexample shows is SAT.
- **Subset:** more `mnist_fc` instances can only make more bugs fire, so the subset may understate full `mnist_fc`; `scripts/run_external_comparison.sh` runs all 90.
- **Thrust 2** (separate self-evaluation card, not the submission): per-verifier timeout prediction AUC is 0.77–0.92 on held-out data and 0.87–0.96 on fresh networks, all above the 0.7 target.